



MODULE NAME:	MODULE CODE:
INFORMATION SYSTEMS 2A	INSY6211/w
SYSTEM ANALYSIS AND DESIGN	SAND6221/p/w

ASSESSMENT TYPE: ASSIGNMENT 1 (PAPER ONLY)

TOTAL MARK ALLOCATION: 100 MARKS

TOTAL HOURS: 10 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. **No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.**
2. **Your assignment must be submitted through TurnItIn.**
3. **Save a copy of your assignment before submitting it.**
4. *Assignments must be typed unless otherwise specified.*
5. *All work must be adequately and correctly referenced.*
6. *This is an individual assignment.*

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where **both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruences: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruences: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

At the end of this specific assignment, students should be able to:

- *LU1 – LO1: Describe the purpose of system analysis and design in the development of information systems;*
- *LU1 – LO2: Describe the characteristics of iterative system development;*
- *LU2 – LO1: Describe the activities of Systems Analysis;*
- *LU2 – LO3: Identify and Understand different kinds of stakeholders and contributions to requirements definition;*
- *LU2 – LO7: Apply the event decomposition technique to identify use cases;*
- *LU2 – LO10: Interpret and create a domain model class diagram;*
- *LU2 – LO11: Develop use case descriptions;*

Assignment Scenario

Read the following scenario. Then answer all questions in this assignment based on the information contained in the scenario. If you make any assumptions when answering a question, please state them clearly.

Scenario: Airport Customer Relations Management System

A local airport has recently upgraded its infrastructure to be able to accommodate international flights. Airport management is expecting an influx of tourists. To ensure the all-round success of this new endeavour, a heavy emphasis is being placed on keeping visitors to the airport as satisfied with the airport and its amenities as possible. To aid in maintaining good customer relations, the airport has decided to invest in the development of a new Customer Relations Management System, which would be accessible online via kiosks strategically placed around the airport as well as a downloadable mobile application.

The new system is envisioned to offer customers the following:

1. Information: updates on flights, airport amenities and services available at the airport;
2. Navigation capability: GPS navigation to various amenities and boarding gates;
3. Instant connection to medical and ambulatory services for any medical emergency.

Question 1		(Marks: 60)
Q.1.1	Conduct the necessary research, then write a report that will detail the role of Systems analysis and Systems Design in the development of the Airport Customer Relations Management System. The report will be presented to the management of the airport in a bid to secure the contract to develop the system and, as such, needs to be structured as follows: 1. Report title (1 Mark). 2. A brief explanation of what Systems Analysis and Design is. Relate your description to the Airport Customer Relations Management System (6 Marks). 3. The importance of conducting Systems Analysis and Design properly for the Airport Customer Relations Management System (10 Marks). 4. The activities that will be conducted as part of Systems Analysis in relation to the development of the Airport Customer Relations Management System (10 Marks). 5. The activities that will be conducted as part of Systems Design in relation to the development of the Airport Customer Relations Management System (10 Marks). 6. Motivation for the use of an iterative approach to the development of the Airport Customer Relations Management System (10 Marks). 7. A conclusion that will not only bring together the contents of the report but will also motivate why you should be selected to develop the new system for the airport (13 Marks). NOTE: Your report should be limited to a maximum of 2 pages.	(60)

Question 2		(Marks: 10)
Stakeholders represent a person or group of people who have an interest in the successful implementation of a new system.		
Q.2.1	Identify five (5) potential stakeholders that you will need to interact with should you be awarded the contract to develop the new Airport Customer Relationship Management System. Also, state how each of these potential stakeholders will contribute to gathering requirements for the new system.	(10)

Question 3		(Marks: 20)
Q.3.1	Using the event decomposition technique, identify any four events and their resulting use cases for the Airport Customer Relations Management System.	(8)
Q.3.2	Create a use case description for any one of the use cases identified in Question 3.1	(12)

Question 4		(Marks: 10)
A passenger at the airport can have multiple flights booked in advance from the given airport.		
Q.4.1	Create a Domain Model Class diagram that will show the classes contained in the statement presented at the beginning of the question. Also, indicate multiplicity as well as any three plausible attributes for each class.	(10)

[TOTAL MARKS: 100]

END OF PAPER